

Chapter 109 — Neuromuscular Dysfunction of the Lower Urinary Tract — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Suprapontine** → detrusor overactivity (DO) + **synergic** sphincters; true DSD never occurs
- **Suprasacral cord** → DO + smooth synergy + **striated detrusor-sphincter dyssynergia (DSD)**
- **Sacral/intrasacral** → areflexia, compliance may fall, fixed striated tone
- **DSD needs lesion between pontine micturition centre (PMC) and sacral cord**
- **Smooth sphincter dyssynergia** if cord lesion $\geq T7-T8$ (column T6)
- **Autonomic dysreflexia**: lesion **above T6**; **first act = drain bladder**; **nitroglycerin 2% paste** above lesion
- **Spinal shock 6–12 weeks** (up to 1–2 years); **clean intermittent catheterisation (CIC) preferred**
- **CIC = gold standard** emptying; volumes **<500 mL**
- **Cauda equina** → **decompress <48 h**
- **Manage by urodynamics, not neurologic level**

OBJECTIVES

- **Five urodynamic parameters**: sensation, detrusor activity, compliance, smooth sphincter, striated sphincter
- **Open bladder neck**: multiple system atrophy (MSA), radical pelvic surgery, myelodysplasia, sacral injury

PATTERNS: NEUROPLASTICITY

- **Bladder outlet obstruction (BOO)**: $\uparrow$ **type III : type I collagen** → $\downarrow$ compliance
- **Afferent plasticity**: nerve growth factor (NGF); **ice-water test = C fibres**
- **Plasticity** explains persisting lower urinary tract symptoms (LUTS) after de-obstruction

DISEASES AT OR ABOVE THE BRAINSTEM

- **Stroke urinary incontinence (UI)**: 32–79% admission → 25–28% discharge → 12–19% later
- **UI persisting 7 days** → poor survival and dependence
- **Commonest UI type post-stroke**: **impaired awareness** — worst prognosis
- **Acute stroke**: retention (areflexia, "cerebral shock"); **chronic**: phasic DO
- **Basal ganglia/thalamus lesion** → voluntary sphincter control preserved
- **"Uninhibited sphincter relaxation"** = misnomer
- **Brainstem**: midbrain none; **pontine** → **low capacity**; **medullary** → emptying failure
- **Dementia**: UI $3\times$ commoner; **anticholinergic** → **10–65% $\uparrow$ dementia risk** → **$\beta 3$ agonist preferred**
- **Traumatic brain injury (TBI)**: above PMC → DO synergic; below PMC → DSD possible
- **Brain tumour**: **superior frontal lobe**; posterior fossa 32–70% dysfunction
- **Normal-pressure hydrocephalus (NPH) triad**: dementia, ataxia, UI (98–100%); **ventriculoperitoneal shunt**
- **Cerebral palsy adults**: detrusor leak point pressure (DLPP) ≥ 40 cm H₂O in $\sim$ half → full work-up
- **Parkinson disease (PD)**: LUTS 35–70%; **storage 50–75%**; **DO commonest**; **gold standard diagnosis = neuropathology (Lewy bodies)**
- **PD**: prevalence 3% ≥ 65 ; 2nd neurodegenerative after Alzheimer
- **PD preferred drugs**: mirabegron, extended-release fesoterodine, trospium, darifenacin
- **Subthalamic deep brain stimulation** improves sensation/capacity

- **MSA = α -synucleinopathy**; MSA-P / MSA-C replace Shy-Drager etc.
- **MSA**: UI + high post-void residual (PVR) + **open bladder neck + sphincter denervation** + erectile dysfunction early
- **MSA**: **outlet surgery** → **worse UI**; desmopressin for nocturia
- **Progressive supranuclear palsy**: vertical gaze palsy; **impaired contraction**; Onuf loss

SPINAL CORD

- **Multiple sclerosis (MS)**: female $\geq 2:1$; LUTS **50–90%**; UI 37–72%; DO 34–99%; DSD 30–65%; areflexia 12–38%; $\sim 10\%$ normal
- **MS presenting LUTS up to 15%** (acute retention/urgency)
- **MS complication risks**: DSD in men, high filling pressure, indwelling catheter
- **MS early**: antimuscarinic + CIC; Credé/Valsalva contraindicated; indwelling → **suprapubic**
- **Spinal cord injury (SCI)**: 54/million/year; **80% male**; mean age 42; **incomplete tetraplegia commonest**
- **Sacral cord at column T12–L1**; **cord ends $\sim L2$**
- **Death now**: pneumonia, septicæmia (not renal)
- **SCI stones**: $\sim 7\%$ in 10 years; **peak first 3 months**; **98% apatite/struvite**
- **Spinal shock**: closed bladder neck; guarding reflex absent; **bulbocavernosus may persist**
- **DSD consequences**: complete > incomplete; continuous > intermittent; **male > female**
- **DSD type 1** → **incomplete**; **types 2, 3** → **complete**; not level
- **Sphincterotomy/sphincter toxin**: only if CIC impossible/refused
- **Low compliance** **<12.5 mL/cm H₂O**; **leak point pressure ≥ 75 cm H₂O** → upper tract damage
- **DO > $\frac{1}{2}$ of filling** → renal damage
- **Thoracolumbar (T10–L2) lesions least predictable**
- **Autonomic dysreflexia (AD)**: **cervical 60% vs thoracic 20%**; ≥ 20 mm Hg systolic rise
- **AD triggers**: bladder/rectal distension; **bradycardia typical**
- **AD prophylaxis**: **terazosin, prazosin**; nifedipine banned in many centres
- **AD intractable**: sacral rhizotomy
- **Vesicoureteral reflux (VUR) in SCI 17–25%**; **first fix pressures, not reflux**
- **Urethral catheter worst** for upper tracts
- **VUR artefact**: capacity $\uparrow$, pressure $\downarrow$ on urodynamics
- **Myelomeningocele**: $\sim 90\%$ of cystica; **1/1000**; LUT dysfunction $\sim 90\%$; **lumbosacral 47%**

- **Myelodysplasia typical:** areflexic + open bladder neck; DSD 10–15%
- **Dysraphic SUI:** bulking safer than sling/artificial urinary sphincter
- **Tethered cord:** 3–15% of myelomeningocele; urinary recovery lags (~4 months)
- **Sensory neurogenic bladder:** tabes, pernicious anaemia/subacute combined degeneration; **motor:** polio

DISTAL TO THE CORD

- **Disc prolapse central** in 1–15%; L4–5, L5–S1; **S2–S4 perineal** sensory loss
- **Disc:** areflexic, normally compliant bladder
- **Cauda equina:** 1–5% of disc prolapses; **somatic recovers before detrusor**
- **Radical pelvic surgery:** abdominoperineal resection 20–68%, radical hysterectomy 16–80%; **permanent 15–20%**
- **Radical pelvic:** open bladder neck + **fixed striated tone** + ↓ compliance; **bethanechol supersensitivity** +
- **Post-pelvic surgery:** CIC and wait 6–12 months
- **Nerve-sparing radical hysterectomy** ↓ LUTS (randomised trial)
- **EPINCONT:** caesarean ↓ moderate–severe UI; **UUI equal**
- **Zoster:** retention days–weeks post-rash; **resolves 1–2 months**
- **Elsberg syndrome:** herpes simplex sacral radiculitis + cerebrospinal fluid pleocytosis
- **Diabetes:** commonest peripheral neuropathy; ↓ sensation → ↑ capacity → ↓ contractility → ↑ PVR
- **Diabetes vs BOO:** pressure–flow distinguishes
- **Diabetes:** timed voiding
- **Guillain–Barré syndrome:** demyelinating 85%; LUT 25–80%; retention 11–30%; **reversible therapy**

MISCELLANEOUS NEUROLOGIC DISEASES

- **AIDS neurogenic LUTS:** ~40% dead within 8 months
- **HTLV-1 myelopathy:** DSD + DO; LUTS up to 60%
- **Schistosomal myelopathy:** DSD → upper tract damage

MISCELLANEOUS RELATED CONDITIONS

- **Untreated DSD:** ~50% of men → complications
- **DLPP ≥40 cm H₂O after sphincterotomy = failure**
- **Blaivas type 1 DSD** → observe unless VUR, hydronephrosis, dysreflexia
- **DSD options:** CIC + DO control (preferred); stents **abandoned**
- **Hinman syndrome** = dysfunctional voiding, no neurology
- **Chronic retention:** PVR ≥300 mL, ≥6 months, ≥2 occasions; long-term → suprapubic
- **Primary bladder neck obstruction (men):** bladder neck incision definitive
- **Trapped prostate** (Turner-Warwick) → small transurethral resection of prostate (TURP)/incision incl. bladder neck
- **Female BOO (Groutz):** Qmax ≤12 mL/s + Pdet@Qmax ≥20 cm H₂O
- **Nitti:** fluoroscopic obstruction; **dysfunctional voiding** commonest cause
- **Prolapse:** test voiding with pelvic organ prolapse reduced before repair
- **Paruresis:** ambulatory urodynamics useful only >40 years
- **Fowler syndrome:** women <30; capacity >1 L, no urge; complex repetitive discharges + decelerating bursts; closure pressure >100 cm H₂O
- **Fowler treatment:** sacral neuromodulation (SNM) (~70%); toxin/hormones fail

- **Post-op retention:** decompress 18–24 h after joint replacement; **phenoxybenzamine** helps; tamsulosin before spine surgery does not
- **Myasthenia gravis:** post-prostatectomy UI ↑
- **Radiation early reaction** peak 4–6 weeks, recovers by 6 months
- **Defunctionalised bladder transplant:** capacity ≥100 mL, voiding pressure ≤100 cm H₂O

EVALUATION, SURVEILLANCE AND TREATMENT

- **Guidelines:** American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU) neurogenic lower urinary tract dysfunction (NLUTD) 2021; European Association of Urology (EAU) Neuro-urology 2022
- **Minimum:** history, examination, urinalysis; **PVR if voiding**
- **Low risk (all):** suprapontine/peripheral, spontaneous void, low PVR, stable → **no routine tests**
- **High risk (any):** poor compliance, ↑ storage pressure, DSD, VUR, hydronephrosis, abnormal renal function
- **Not low risk = unknown** until urodynamics + imaging + renal function
- **Defer assessment during spinal shock/acute brain injury**
- **Chronic catheter:** suprapubic > urethral
- **Hydrophilic catheters** ↓ urinary tract infection (UTI)
- **Neurogenic DO:** drugs first-line → onabotulinumtoxinA
- **Posterior tibial nerve stimulation (PTNS):** spontaneous voiders; **SNM:** NLUTD without SCI/spina bifida
- **UTI:** treat only symptomatic; ≥5 days; 7–14 relapse; **no prophylaxis with CIC/indwelling**
- **Bladder cancer:** risk = catheter, stones, UTI; often muscle-invasive
- **AUA/SUFU:** no screening cystoscopy
- **AUA/SUFU follow-up (moderate/high):** yearly renal function; imaging every 1–2 years
- **Bowel reconstruction:** annual electrolytes, renal function, imaging
- **American Paraplegia Society:** annual 5–10 years; **renal plasma flow fall >20% → investigate**
- **EAU:** dipstick 2-monthly; **ultrasound 6-monthly;** specialist review 1–2 years

CLASSIC TRAPS

- **Trap:** EMG flare in stroke/PD = DSD — **Actually:** pseudodysynergia
- **Trap:** TURP contraindicated in PD — **Actually:** MSA; confirm PD first
- **Trap:** Neurologic level predicts bladder — **Actually:** urodynamics required
- **Trap:** Areflexia excludes suprasacral lesion — **Actually:** occurs in ~¼
- **Trap:** MS upper tracts behave like SCI — **Actually:** rarely damaged
- **Trap:** VUR must be corrected first — **Actually:** lower bladder pressure first
- **Trap:** Upper tracts safe without VUR — **Actually:** high pressure alone damages
- **Trap:** Prostatectomy for post-pelvic-surgery retention — **Actually:** only if BOO proven
- **Trap:** Diabetes = hypocontractile only — **Actually:** DO most frequent finding
- **Trap:** Catheterised NLUTD need screening cystoscopy — **Actually:** not recommended